

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027****Vision:** Dream of where you want.**Mission:** Means to achieve Vision**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)**Keywords of POs:**

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab: Hybrid Computing Systems
Semester	7	Course Code	23IOT1704
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	02
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	Creation and Management of Virtual Machines using Cloud Platforms
Theory (100 words)	A Virtual Machine (VM) is a virtualized computing environment that operates independently within a physical computer or cloud server. It uses a virtualization layer called a hypervisor to obtain resources such as CPU, RAM, storage, and networking from the underlying hardware. Each VM can have its own operating system and applications, allowing different operating systems to run on the same physical machine. In cloud computing, VMs can be provisioned remotely according to the user's requirements. Parameters such as operating system, processor capacity, memory, storage, and network configuration can be selected during VM creation. VMs can also be started, stopped, restarted, resized, or deleted as needed. Virtualization improves hardware utilization by allowing multiple independent VMs to share a single physical server. Cloud providers such as AWS, Microsoft Azure, and Google Cloud offer VM-based computing services. VMs are widely used for software development, testing, server hosting, application deployment, and isolated computing environments.



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Procedure and Execution (100 Words)	Implementation Steps: 1. Open a web browser and visit the AWS Management Console. 2. Sign in to the AWS account using the required credentials. 3. Search for and open the EC2 (Elastic Compute Cloud) service. 4. Select a suitable AWS Region from the top-right corner. 5. Click on "Launch Instance" to create a new Virtual Machine. 6. Enter a name for the Virtual Machine, such as "My-First-VM". 7. Under "Application and OS Images", select Ubuntu Server. 8. Select a suitable instance type that is eligible for the applicable AWS Free Tier, if available. 9. Create or select a Key Pair and download the key pair file. 10. Under Network Settings, allow SSH (Port 22) for connecting to the Linux Virtual Machine. 11. Review the storage and other settings. 12. Click on "Launch Instance". 13. Open the "Instances" section and check the created Virtual Machine. 14. Wait until the instance state shows "Running". 15. Select the instance and click on "Connect". 16. Select an available connection method, such as EC2 Instance Connect. 17. Click on "Connect" to open the Virtual Machine terminal. 18. Run a basic command such as "uname -a" or "ls" to verify that the Virtual Machine is working. 19. Return to the EC2 dashboard and use options such as Stop Instance and Start Instance to manage the Virtual Machine. 20. After completing the practical, terminate the instance if it is no longer required to avoid unnecessary charges.



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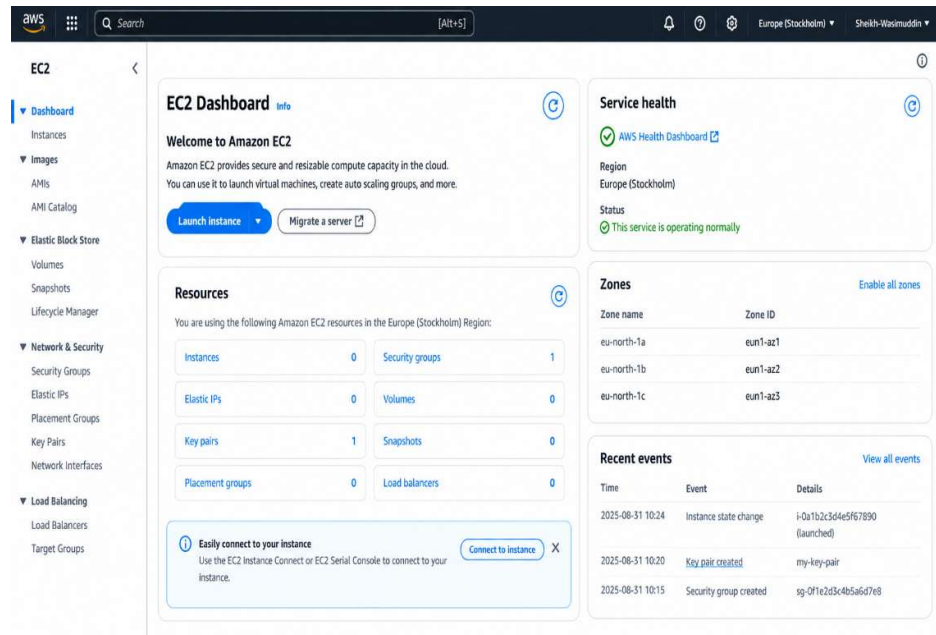
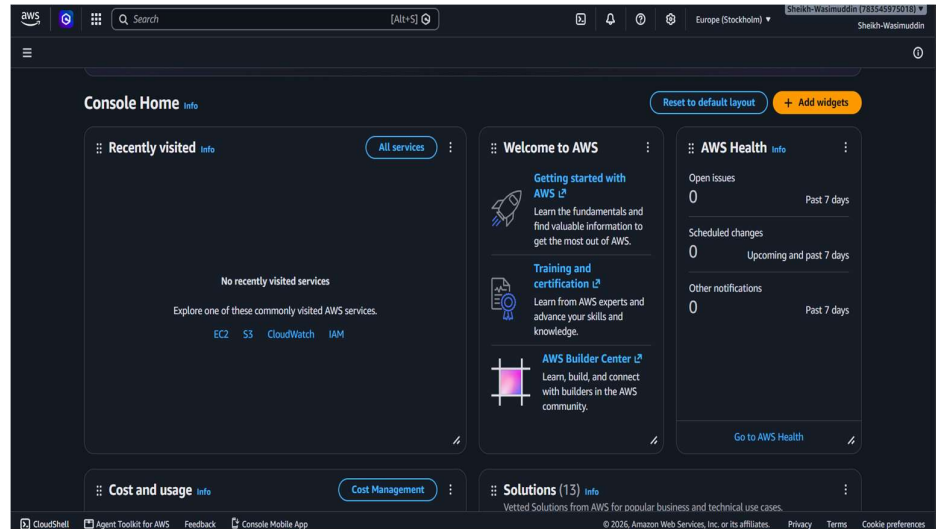
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Stepwise Screenshots with steps:





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EC2

Dashboard

Instances

Images

Elastic Block Store

Network & Security

Load Balancing

Launch instance

1 Choose AMI

2 Choose instance type

3 Configure instance

4 Add storage

5 Add tags

6 Review and launch

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, with common operating systems and software pre-installed.

Application and OS images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance.

Quickstart

My AMIs

Shared with me

AWS Marketplace

Community AMIs

Ubuntu 22.04 LTS

Amazon Linux 2

Red Hat 9

SUSE Linux 15 SP5

Windows Windows Server 2022

macOS macOS Big Sur

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type

Free tier eligible

64-bit (x86)

Select

Summary

Number of instances

1

Software image (AMI)

Canonical, Ubuntu, 22.04 LTS, ...

Instance type

t2.micro

Free tier eligible

Key pair (login)

Not selected

Edit

Network settings

VPC - vpc-0e2b3c1f4d5e6a7b8

Subnet - No preference

Edit

Storage (volumes)

1 volume(s) - 8 GiB

Edit

Next: Choose instance type

EC2

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Application and OS images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance.

Quickstart

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Ubuntu 22.04 LTS

Amazon Linux 2

Red Hat 9

SUSE Linux 15 SP5

Windows Windows Server 2022

macOS macOS Big Sur

Amazon Machine Image (AMI) Details

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type

ami-0c7217cd6d317cfc7 (64-bit (x86))

Ubuntu Server 22.04 LTS (HVM), EBS General Purpose (gp3) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Instance type

Choose an instance type that meets your compute, memory, and storage needs.

Filter instance types

Instance type	vCPUs	Memory (GiB)	Storage (GiB)	Network performance	On-Demand hourly rate
t2.micro	1	1	EBS only	Low to Moderate	\$0.00

Summary

Number of instances

1

Software image (AMI)

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type

Instance type

t2.micro

Free tier eligible

Key pair (login)

Not selected

Edit

Network settings

VPC - vpc-0e2b3c1f4d5e6a7b8

Subnet - No preference

Edit

Security groups

Select existing security group

Edit

Storage (volumes)

1 volume(s) - 8 GiB

Edit

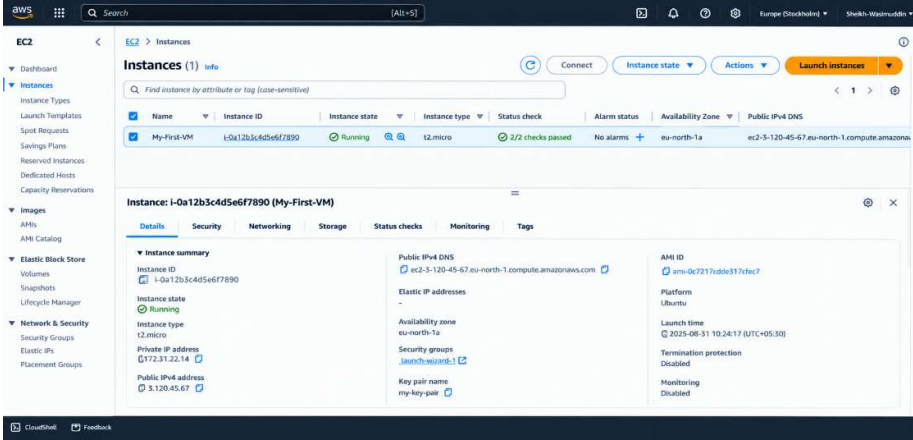
Next: Configure instance

CloudShell

Feedback

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Conclusion	The practical successfully demonstrated the basic concepts of Cloud Computing, its architecture, service models, and deployment models. The roles of the Front End, Back End, and Internet in cloud architecture were studied. The three service models, IaaS, PaaS, and SaaS, were understood based on their control and management responsibilities. The four deployment models, Public, Private, Hybrid, and Community Cloud, were also studied based on their accessibility, ownership, security, and usage. Thus, the practical provided a clear understanding of how cloud services are organized and delivered according to different user and organizational requirements.
Date	03/07/2026